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Johannes Kepler

IN KEPLER'S LETTERS we find ourselves confronted with a sensitive personality, passionately devoted to the quest for deeper insight into the character of natural processes—a man who reached the exalted goal he set himself in spite of all internal and external difficulties. Kepler's life was devoted to the solution of a dual problem. The sun and the planets change their apparent position with reference to their background of fixed stars in a complex manner open to immediate observation. In other words, all the observations and records compiled with such care dealt not actually with the movements of the planets in space but with temporal shifts undergone by the direction earth-planet in the course of time.

Once Copernicus had convinced the small group capable of grasping it that in this process the sun must be regarded as being at rest, with the planets, including the earth, revolving about the sun, the first great problem proved to be this: to determine the true motions of the planets, including the earth, as they might be visible to an observer on the nearest fixed star who was equipped with a perfect stereoscopic double-telescope. This was Kepler's first great problem. The second problem was embodied in this question: What are the mathematical laws under which these motions proceed? It is plain that the solution of the second problem, if at all within reach of the human mind, was predicated on the solution of the first. Before a theory explaining a certain process can be tested, that process must first be known.

Kepler's solution of the first problem is based on a truly inspired notion that made possible the determination of the true orbit of the earth. To construct that orbit, a second fixed point in planetary space, in addition to the sun, is required. When such a second point is available, it and the sun may both be used as points of reference for angular measurements, and the earth's true orbit can be determined by the same methods of triangulation that customarily serve in surveying and cartography.

But where was such a second fixed point to be found, since all visible objects, except the sun, themselves execute motions that are not known in detail? This was Kepler's answer: The apparent motions of the planet Mars are known with great accuracy, including the time of its revolution about the sun (the "Martian year"). It is probable that at the end of each Martian year Mars is at the same spot in (planetary) space. If we limit ourselves for the time being to these points in time, then the planet Mars represents for them a fixed point in planetary space, a point that may be used in triangulation.

Employing this principle, Kepler first of all determined the true motion of the earth in planetary space. Since the earth itself may be used as a point for triangulation at any time, he was also able to determine the true motions of the other planets from his observations.

This is how Kepler gained the basis for formulating the three fundamental laws with which his name will remain associated for all time to come. Today, after the fact, no one can fully appreciate how much ingenuity, how much hard and tireless work was required to discover these laws and ascertain them with such precision.

The reader ought to know this as he learns from the letters under what hardships Kepler accomplished this gigantic work. He refused to be paralyzed or discouraged either by poverty or by the lack of comprehension among those of his contemporaries who had the power to shape his life and work. Yet he was dealing with a subject that offered immediate danger to him who professed the truth. But Kepler was one of the few who are simply incapable of doing anything but stand up openly for their convictions in every field. At the same time he was not one who took undiluted pleasure in personal controversy, as was plainly the case with Galileo, whose inspired barbs delight the informed reader even today. Kepler was a devout Protestant, but he made no secret of the fact that he did not approve of all decisions by the Church. He was, accordingly, regarded as a kind of moderate heretic and treated as such.

This brings me to the inner difficulties Kepler had to overcome—difficulties at which I have already hinted. They are not as readily perceived as the outward difficulties. Kepler's lifework was possible only once he succeeded in freeing himself to a great extent of the intellectual traditions into which he was born. This meant not merely the religious tradition, based on the authority of the Church, but general concepts on the nature and limitations of action within the universe and the human sphere, as